

Deep Learning-Based Classification of Medial Deltoid EMG Signals for Intramuscular Injection Guidance

Final Project Report

Submitted by:

Emad Alabdul Rahman
Student Number: 241307126

Supervised by:

Assoc.Prof. MUSTAFA HİKMET BİLGEHAN UÇAR

Kocaeli University
Technology Faculty
Information Systems Engineering

December 23, 2025

Abstract

This study presents a deep learning-based approach for classifying electromyography (EMG) signals from different regions of the deltoid muscle, with particular focus on identifying the medial deltoid region used for intramuscular injections. The deltoid muscle consists of three distinct portions: anterior, medial, and posterior. Accurate identification of the medial deltoid is clinically important for safe vaccine administration and intramuscular injections. Using EMG data from the Zenodo repository (record 15645794), we developed a one-dimensional convolutional neural network (1D CNN) that processes windowed EMG signals to distinguish medial deltoid activity from other deltoid regions. The dataset comprised recordings from three deltoid regions (DELT1, DELT2, DELT3) which were preprocessed using band-pass filtering (20-450 Hz), full-wave rectification, and low-pass filtering (6 Hz) to extract the EMG envelope. A sliding window approach with window size of 50 samples was employed to capture temporal patterns in the signals. The trained model achieved a test accuracy of 83.91 percent, with an ROC-AUC score of 0.9397 and average precision of 0.9015. For the medial deltoid class (DELT2), the model demonstrated a recall of 85.98 percent and precision of 71.50 percent, indicating strong capability in identifying target muscle activity. These results suggest that EMG-based deep learning systems can provide reliable support for clinical applications requiring precise muscle region identification, with potential applications in training simulators and injection guidance systems.

Contents

1	Introduction	4
2	Dataset Description	5
3	Methodology	6
3.1	Overview of the Processing Pipeline	6
3.2	Data Extraction	6
3.3	Signal Preprocessing and Filtering	7
3.4	Data Restructuring and Labeling	9
3.5	Windowing Approach	10
3.6	Train-Validation-Test Split	10
3.7	Model Architecture	11
3.8	Class Imbalance Handling	11
3.9	Model Training	12
4	Results	13
4.1	Model Performance Metrics	13
4.2	Class-Wise Performance Analysis	13
4.3	Confusion Matrix Analysis	14
4.4	Performance Visualizations	14
4.5	Class Distribution Analysis	16
5	Discussion	16
6	Conclusion	18

List of Figures

1	Anatomical illustration showing the three portions of the deltoid muscle: anterior, medial, and posterior regions.	5
2	Complete methodology flowchart showing the data processing pipeline from raw MATLAB files to final model evaluation. The pipeline includes five main stages: Data Acquisition for extracting deltoid channels, Signal Preprocessing for filtering and envelope extraction, Data Preparation for windowing and dataset splitting, Model Training with weighted loss optimization, and Evaluation on the test set.	6
3	Raw EMG signals from the three deltoid muscle channels (DELT1, DELT2, DELT3) showing the unprocessed electrical activity patterns. .	8
4	Filtered EMG envelope signals after band-pass filtering, rectification, and low-pass filtering, showing the smooth muscle activation patterns. .	8
5	Comparison of raw (light lines) and filtered (dark lines) EMG signals for each deltoid channel, demonstrating the effectiveness of the preprocessing pipeline.	9
6	Comprehensive evaluation plots: (top-left) confusion matrix heatmap showing prediction patterns, (top-right) ROC curve demonstrating classification performance across thresholds, (bottom-left) precision-recall curve showing the trade-off between precision and recall, (bottom-right) distribution of predicted probabilities for each true class showing model confidence.	15

1 Introduction

Electromyography (EMG) represents a fundamental technique for measuring electrical activity produced by skeletal muscles during contraction. In clinical practice, accurate identification of specific muscle regions plays a critical role in numerous procedures, particularly intramuscular injections and vaccine administration. The deltoid muscle, located in the shoulder region, serves as one of the primary sites for intramuscular injections due to its accessibility and relatively large muscle mass. However, the deltoid muscle is not a homogeneous structure; it consists of three anatomically and functionally distinct portions: the anterior deltoid, the medial (or middle) deltoid, and the posterior deltoid.

The medial deltoid portion represents the preferred target for intramuscular injections because of its optimal depth, muscle mass, and distance from major neurovascular structures. Incorrect needle placement, either too superficial or into adjacent muscle regions, can result in reduced vaccine efficacy, increased patient discomfort, or potential injury to surrounding tissues. Traditional training methods for healthcare professionals rely on anatomical landmarks and palpation techniques, which can vary in reliability depending on patient body habitus and clinician experience.

Recent advances in machine learning and deep learning have opened new possibilities for analyzing physiological signals and extracting clinically relevant patterns. Convolutional neural networks, originally developed for image processing tasks, have demonstrated remarkable success in processing time-series data such as EMG signals. The one-dimensional variant of CNNs (1D CNN) has proven particularly effective for EMG signal analysis due to its ability to automatically learn relevant features from raw or preprocessed signals without requiring extensive manual feature engineering.

This project aims to develop an automated classification system capable of distinguishing EMG signals originating from the medial deltoid region from those generated by the anterior and posterior deltoid portions. By leveraging deep learning techniques and a publicly available EMG dataset, we seek to create a model that could eventually support clinical training applications, injection guidance systems, or real-time muscle activity monitoring during medical procedures.

The specific objectives of this research include: (1) extracting and preprocessing EMG data from multiple deltoid muscle regions, (2) implementing appropriate signal processing techniques to enhance signal quality and extract relevant features, (3) developing a 1D convolutional neural network architecture optimized for EMG signal classification, (4) evaluating model performance using standard classification metrics, and (5) assessing the clinical feasibility of such a system for supporting intramuscular injection procedures.

2 Dataset Description

The dataset used in this study was obtained from the Zenodo repository under record number 15645794. This publicly available dataset contains EMG recordings from multiple shoulder muscles captured during dynamic movement trials. The recordings were performed using surface EMG electrodes placed over various muscle groups in the shoulder region, with a sampling frequency of 1000 Hz.

For this project, we focused exclusively on the three deltoid muscle channels, which correspond to the three anatomical portions of the deltoid muscle. The EMG channels were labeled as follows: DELT1 representing the anterior deltoid, DELT2 representing the medial deltoid (the target class for our classification task), and DELT3 representing the posterior deltoid. The anatomical arrangement of these muscle portions is illustrated in Figure 1.

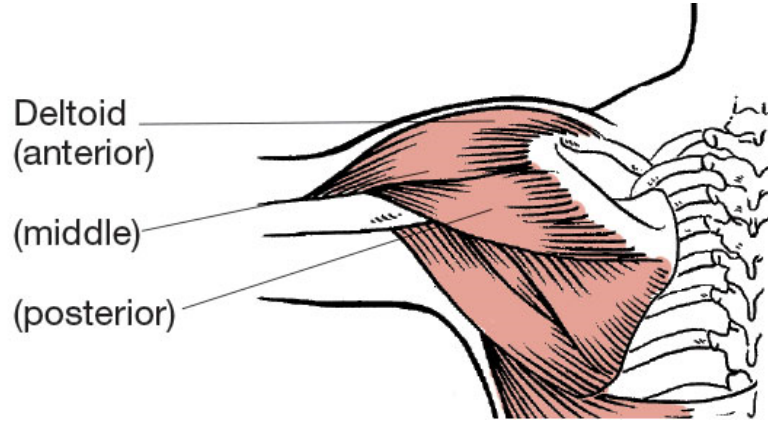


Figure 1: Anatomical illustration showing the three portions of the deltoid muscle: anterior, medial, and posterior regions.

The dataset consisted of multiple recording sessions from a single participant (H05), including calibration trials and dynamic movement trials. Each trial was stored as a MATLAB file (AnalogData.mat) containing structured data with raw EMG signals and associated metadata. The calibration trials (H005dynamicCal06 through H005dynamicCal18) provided baseline recordings, while the dynamic trials (H005dynamictrial01 through H005dynamictrial55) captured muscle activity during various shoulder movements.

After extracting the deltoid-specific channels from all available trials, the combined dataset comprised 1,009,002 individual time points across all three deltoid channels. This substantial dataset size provided sufficient examples for training a deep learning model while allowing for proper validation and testing procedures. The distribution of recording lengths varied across trials, with individual trials containing between 7,746 and 42,866 samples, reflecting the natural variation in movement execution duration.

The class distribution in the dataset exhibited imbalance, as expected in physiolog-

ical recordings where different muscle portions may be activated to different degrees depending on the movement being performed. After the windowing procedure (described in Section 3.5), the dataset contained approximately 66.67 percent samples from the combined anterior and posterior deltoid regions (Class 0) and 33.33 percent samples from the medial deltoid region (Class 1). This imbalance was addressed during model training through the use of weighted loss functions, as detailed in Section 3.9.

3 Methodology

3.1 Overview of the Processing Pipeline

Figure 2 presents a comprehensive flowchart illustrating the complete data processing and model development pipeline employed in this study. The workflow begins with data extraction from the Zenodo repository and progresses through multiple preprocessing stages, windowing, model training, and evaluation. Each major step is described in detail in the following subsections.

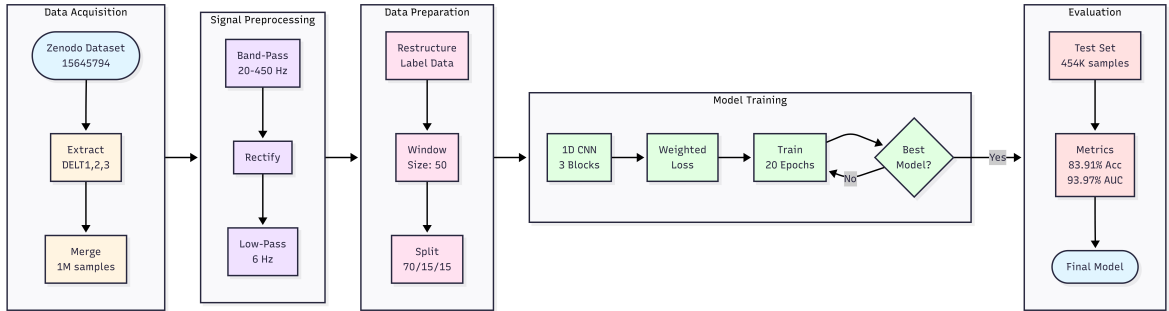


Figure 2: Complete methodology flowchart showing the data processing pipeline from raw MATLAB files to final model evaluation. The pipeline includes five main stages: Data Acquisition for extracting deltoid channels, Signal Preprocessing for filtering and envelope extraction, Data Preparation for windowing and dataset splitting, Model Training with weighted loss optimization, and Evaluation on the test set.

3.2 Data Extraction

The first stage of the data processing pipeline involved extracting the relevant EMG channels from the MATLAB data files. We developed a Python-based extraction tool (EMGExtractor class) that systematically traversed the directory structure containing all trial folders, located each AnalogData.mat file, and extracted the three deltoid muscle channels (EMG7, EMG8, and EMG9, corresponding to DELT1, DELT2, and DELT3 respectively).

The extraction process included an optional preprocessing step to remove time points where all three channels simultaneously registered zero values, which typically

indicated periods of no recording or electrode disconnection. This step helped reduce the dataset size and eliminate non-informative data points. Each trial was saved as a separate CSV file containing the three deltoid channels, and subsequently all individual trial files were merged into a single comprehensive dataset (ALL_DELT_RAW.csv) for subsequent processing stages.

The extraction procedure successfully processed 68 individual trials, producing a merged raw dataset containing 1,009,002 samples. This consolidation step facilitated efficient data handling in subsequent processing stages and ensured consistent data formatting throughout the analysis pipeline.

3.3 Signal Preprocessing and Filtering

Raw EMG signals inherently contain various sources of noise and artifacts, including powerline interference, motion artifacts, electrode noise, and physiological signals from other sources. To extract meaningful muscle activity information, we implemented a comprehensive signal preprocessing pipeline based on established EMG processing techniques.

The preprocessing procedure consisted of three sequential filtering stages. First, a fourth-order Butterworth band-pass filter with cutoff frequencies of 20 Hz and 450 Hz was applied to isolate the frequency range containing the majority of EMG signal power while removing low-frequency motion artifacts and high-frequency noise. This frequency band encompasses the typical spectral content of surface EMG signals from voluntary muscle contractions.

Following band-pass filtering, full-wave rectification was performed by taking the absolute value of the filtered signal. This step is essential for EMG envelope extraction, as it converts the bipolar EMG signal (which oscillates around zero) into a unipolar signal that better represents the underlying muscle activation level.

The third filtering stage involved applying a fourth-order Butterworth low-pass filter with a cutoff frequency of 6 Hz to the rectified signal. This low-pass filtering operation smoothed the rectified EMG, producing an envelope that represents the time-varying amplitude of muscle activity. The 6 Hz cutoff frequency was selected based on the typical rate of force modulation during voluntary movements, which generally falls below this frequency.

Figure 3 presents the raw EMG signals from all three deltoid channels before pre-processing, illustrating the high-frequency oscillatory nature of the raw recordings.

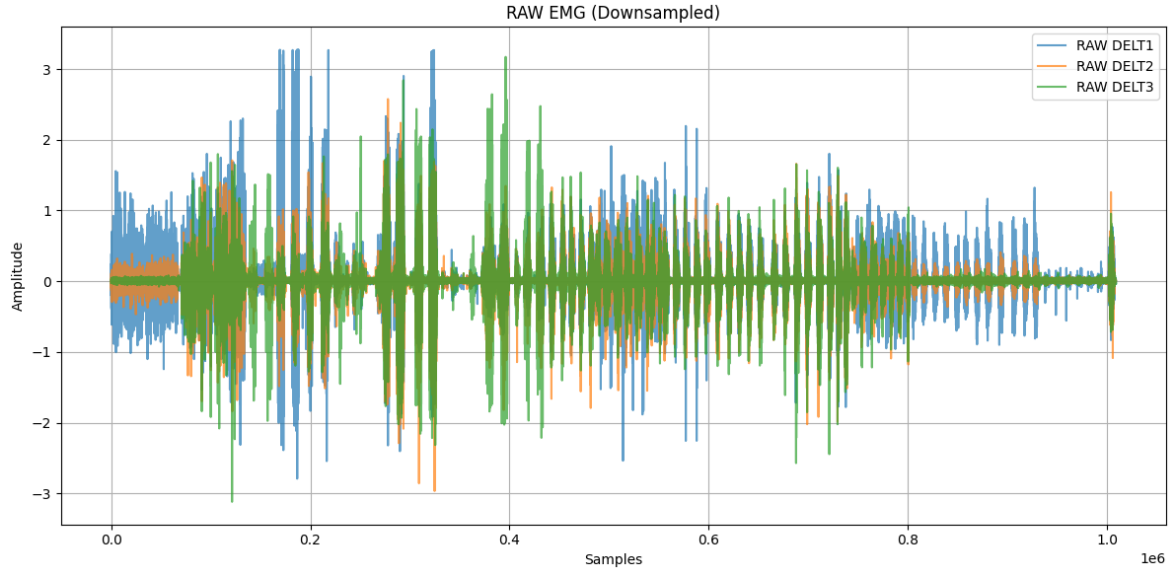


Figure 3: Raw EMG signals from the three deltoid muscle channels (DELT1, DELT2, DELT3) showing the unprocessed electrical activity patterns.

Figure 4 shows the corresponding filtered EMG envelopes after applying the complete preprocessing pipeline, demonstrating the effective noise reduction and envelope extraction achieved through the filtering procedure.

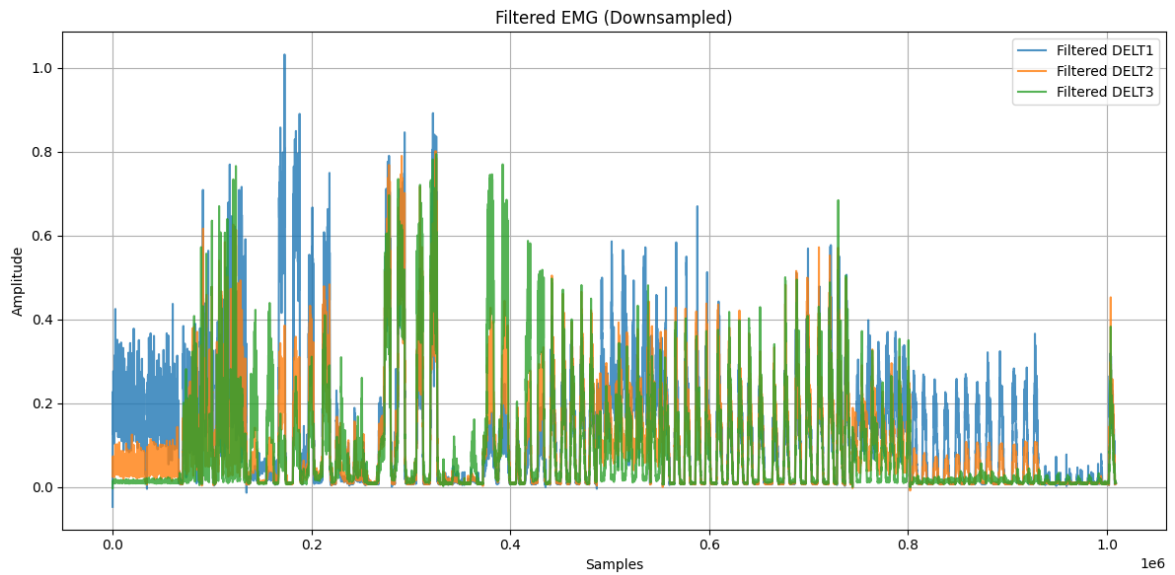


Figure 4: Filtered EMG envelope signals after band-pass filtering, rectification, and low-pass filtering, showing the smooth muscle activation patterns.

Figure 5 provides a direct comparison between raw and filtered signals for each deltoid channel, clearly illustrating the signal enhancement achieved through preprocessing.

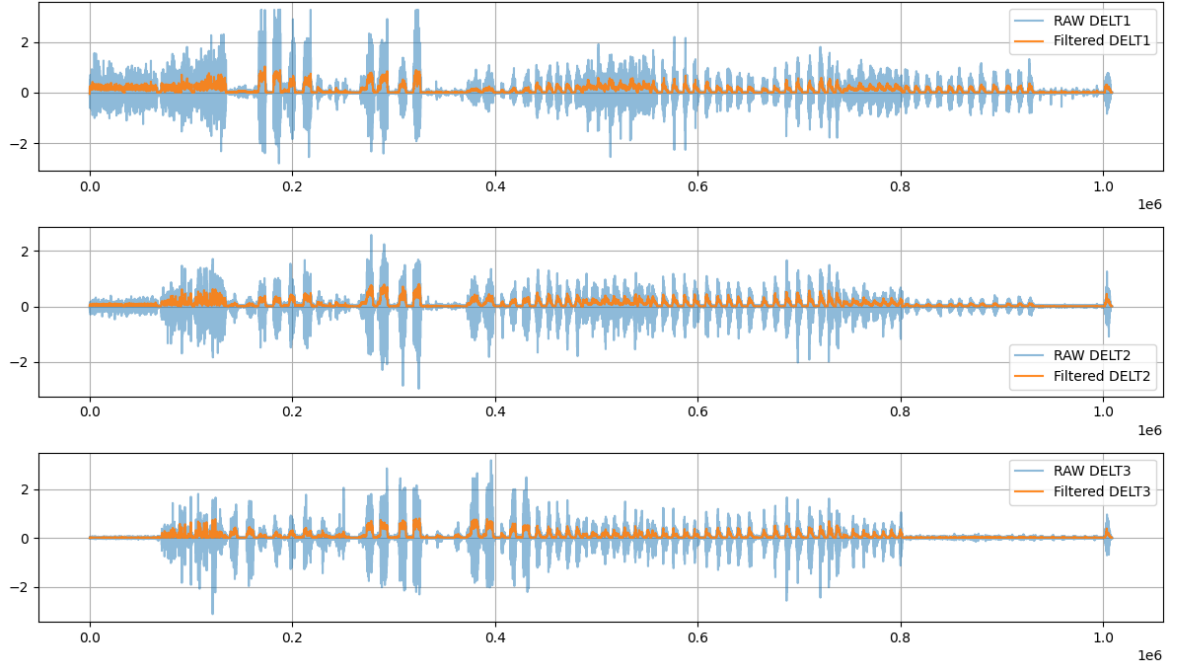


Figure 5: Comparison of raw (light lines) and filtered (dark lines) EMG signals for each deltoid channel, demonstrating the effectiveness of the preprocessing pipeline.

All filtering operations were implemented using the `scipy.signal` library in Python, employing zero-phase filtering (`filtfilt` function) to avoid introducing phase distortion that could affect the temporal alignment of signals from different channels.

3.4 Data Restructuring and Labeling

To prepare the preprocessed EMG data for classification, we restructured the dataset from its original wide format (where each row contained simultaneous recordings from all three channels) into a long format suitable for supervised learning. This restructuring process involved creating individual samples for each channel at each time point.

For each original data point in the filtered dataset, three separate samples were generated: one for DELT1, one for DELT2, and one for DELT3. Each sample consisted of the EMG amplitude value and a binary label indicating whether it originated from the medial deltoid (DELT2, labeled as 1) or from one of the other deltoid portions (DELT1 or DELT3, labeled as 0). This binary classification framework aligned with our clinical objective of specifically identifying medial deltoid activity.

The restructured dataset thus contained three times as many samples as the original filtered data, with each sample representing a single-channel EMG value at a specific time point. This format enabled the subsequent windowing procedure and facilitated the binary classification approach.

3.5 Windowing Approach

Rather than treating each individual time point as an independent sample, we adopted a sliding window approach to capture temporal context and sequential patterns in the EMG signals. This windowing strategy is motivated by the physiological understanding that muscle activation patterns evolve over time, and single time points in isolation provide limited information about the underlying muscle activity state.

We implemented a centered sliding window with a window size of 50 samples. For each position in the signal (excluding the initial and final 25 samples to ensure complete windows), we extracted a window of 50 consecutive EMG values centered at that position. The label for each window was determined by the label of the center sample, following the assumption that the window predominantly represents the muscle activity state at its center point.

This windowing procedure transformed the dataset from individual time point samples into windowed sequences. Each window represented 50 milliseconds of EMG activity (given the 1000 Hz sampling rate), a duration sufficient to capture meaningful temporal patterns while remaining short enough to assume relatively stable muscle activation within the window.

The sliding window approach with unit stride resulted in a large number of overlapping windows, which served to augment the effective training data and helped the model learn robust representations of EMG patterns. After windowing, the dataset contained 454,044 test samples (with proportional numbers allocated to training and validation sets), each representing a 50-sample sequence of EMG values.

3.6 Train-Validation-Test Split

To ensure rigorous evaluation of model performance and avoid overfitting, we partitioned the windowed dataset into three subsets: training, validation, and test sets. The initial split allocated 70 percent of the data to training and 30 percent to a temporary set. This temporary set was then equally divided into validation and test sets, each comprising 15 percent of the total data.

All splits were performed using stratified sampling to maintain the class distribution across subsets. This stratification ensured that both the minority class (medial deltoid) and the majority class (other deltoid portions) were proportionally represented in each subset, preventing bias in model training and evaluation.

The final dataset allocation consisted of training set with a substantial number of samples for model optimization, validation set for hyperparameter tuning and early stopping, and test set reserved exclusively for final performance evaluation. This three-way split provided robust estimates of model generalization performance while preventing information leakage between training and evaluation.

3.7 Model Architecture

We developed a one-dimensional convolutional neural network architecture specifically designed for processing windowed EMG signals. The model architecture consisted of three main components: a feature extraction module, a global pooling layer, and a classification head.

The feature extraction module comprised three convolutional blocks, each containing a 1D convolutional layer, batch normalization, ReLU activation, and max pooling. The first block used 32 filters with kernel size 7, the second block employed 64 filters with kernel size 5, and the third block utilized 128 filters with kernel size 3. This progressive increase in the number of filters and decrease in kernel size follows a common design pattern in deep learning, where early layers capture broad patterns and deeper layers extract more refined features.

Batch normalization layers were incorporated after each convolutional operation to stabilize training, accelerate convergence, and provide mild regularization. Max pooling with pool size 2 was applied after each block to reduce the temporal dimension and extract the most salient features from each local region.

Following the convolutional blocks, a global average pooling layer reduced each feature map to a single value, producing a fixed-length feature vector regardless of the input sequence length. This architectural choice provided several advantages, including parameter reduction, translation invariance, and elimination of the need to specify a fixed input length.

The classification head consisted of two fully connected layers. The first layer mapped the 128-dimensional feature vector to 128 hidden units with ReLU activation, followed by a dropout layer with probability 0.5 for regularization. The final layer produced a single logit value for binary classification, which was passed through a sigmoid function during inference to obtain class probabilities.

The complete model contained 89,217 trainable parameters, a modest number that helped prevent overfitting while providing sufficient capacity to learn the relevant EMG patterns. The model architecture can be summarized as follows: Input (1 channel, 50 samples) $\rightarrow$ Conv1D(32, k=7) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool(2) $\rightarrow$ Conv1D(64, k=5) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool(2) $\rightarrow$ Conv1D(128, k=3) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool(2) $\rightarrow$ GlobalAvgPool $\rightarrow$ Dense(128) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(1) $\rightarrow$ Output.

3.8 Class Imbalance Handling

The binary classification problem exhibited class imbalance, with approximately twice as many samples from the combined anterior and posterior deltoid regions (Class 0) compared to the medial deltoid region (Class 1). Without specific handling, this im-

balance would bias the model toward predicting the majority class, resulting in poor performance on the clinically important minority class.

To address this imbalance, we employed a weighted binary cross-entropy loss function during training. The positive class weight was computed as the ratio of negative samples to positive samples, which yielded a value of approximately 2.0. This weighting scheme penalized misclassification of medial deltoid samples more heavily than misclassification of other deltoid samples, encouraging the model to learn robust representations of the minority class despite its lower frequency in the training data.

The weighted loss function provided an effective mechanism for balancing model attention between classes without requiring data resampling techniques such as oversampling or undersampling, which can introduce their own biases and computational inefficiencies.

3.9 Model Training

Model training was performed using the Adam optimizer with an initial learning rate of 0.001. The Adam optimizer was selected for its adaptive learning rate properties and robust performance across a wide range of deep learning tasks. No learning rate scheduling was employed, as preliminary experiments indicated stable convergence with a fixed learning rate.

Training proceeded for 20 epochs with batch size 256. During each epoch, the entire training set was processed in randomized order, with gradient updates performed after each batch. After each training epoch, the model was evaluated on the validation set to monitor training progress and identify potential overfitting.

The best model was selected based on validation loss, with the model state from the epoch achieving the lowest validation loss preserved for final evaluation. This early stopping criterion prevented overfitting to the training data and ensured that the final model represented the optimal balance between training performance and generalization capability.

Training was performed on CUDA-enabled GPU hardware when available, significantly accelerating the optimization process. The complete training procedure, including data loading, forward passes, backward propagation, and validation, required approximately 4 minutes per epoch on the available hardware.

Throughout training, we monitored multiple metrics including training loss, validation loss, and validation accuracy to ensure proper model convergence and identify any training instabilities or performance plateaus.

4 Results

4.1 Model Performance Metrics

The trained 1D CNN model was evaluated on the held-out test set comprising 454,044 windowed samples. Table 1 presents the fundamental performance metrics achieved by the model.

Table 1: Basic performance metrics on the test set

Metric	Value
Test Loss	0.4012
Test Accuracy	83.91%
ROC-AUC Score	0.9397
Average Precision	0.9015

The model achieved an overall accuracy of 83.91 percent, indicating correct classification of approximately five out of every six test samples. More importantly, the ROC-AUC score of 0.9397 demonstrated excellent discriminative ability, substantially exceeding the baseline value of 0.5 that would be expected from random classification. The high average precision score of 0.9015 further confirmed the model’s strong performance, particularly relevant given the class imbalance in the dataset.

4.2 Class-Wise Performance Analysis

Table 2 provides detailed performance metrics broken down by class, revealing important insights into the model’s behavior for each deltoid muscle category.

Table 2: Class-wise performance metrics

Class	Precision	Recall	F1-Score
Class 0 (DELT1, DELT3)	92.20%	82.87%	87.29%
Class 1 (DELT2)	71.50%	85.98%	78.08%

For Class 0 (anterior and posterior deltoid combined), the model achieved precision of 92.20 percent, indicating that when the model predicted a sample belonged to this class, it was correct 92.20 percent of the time. The recall of 82.87 percent showed that the model successfully identified 82.87 percent of all actual Class 0 samples. The resulting F1-score of 87.29 percent represented a strong harmonic mean of precision and recall.

For Class 1 (medial deltoid, the clinically important target), the model demonstrated recall of 85.98 percent, meaning it successfully identified nearly 86 percent

of all medial deltoid activity. The precision of 71.50 percent indicated that approximately 71.50 percent of samples predicted as medial deltoid were indeed from that muscle region. While lower than the precision for Class 0, this performance level remained clinically relevant, particularly given the high recall value. The F1-score of 78.08 percent reflected the balance between these metrics.

The specificity for Class 0 was 82.87 percent (equivalent to the recall), while the sensitivity for Class 1 reached 85.98 percent. These values indicated that the model performed reasonably well at correctly identifying both positive and negative cases, though with slightly better sensitivity for detecting medial deltoid activity.

4.3 Confusion Matrix Analysis

The confusion matrix presented in Table 3 provides complete information about the model’s predictions across both classes.

Table 3: Confusion matrix for test set predictions

	Predicted 0	Predicted 1	Total
Actual 0	250,836	51,860	302,696
Actual 1	21,217	130,131	151,348
Total	272,053	181,991	454,044

The confusion matrix revealed that the model correctly classified 250,836 samples as Class 0 (true negatives) and 130,131 samples as Class 1 (true positives). The model produced 51,860 false positives (Class 0 samples incorrectly classified as Class 1) and 21,217 false negatives (Class 1 samples incorrectly classified as Class 0).

From a clinical perspective, the relatively low false negative rate was particularly encouraging, as these errors represented instances where medial deltoid activity was present but not detected by the model. The false positive rate, while higher, represented a more conservative type of error where the model indicated possible medial deltoid activity when other deltoid regions were active.

4.4 Performance Visualizations

Figure 6 presents comprehensive visualizations of model performance across multiple dimensions.

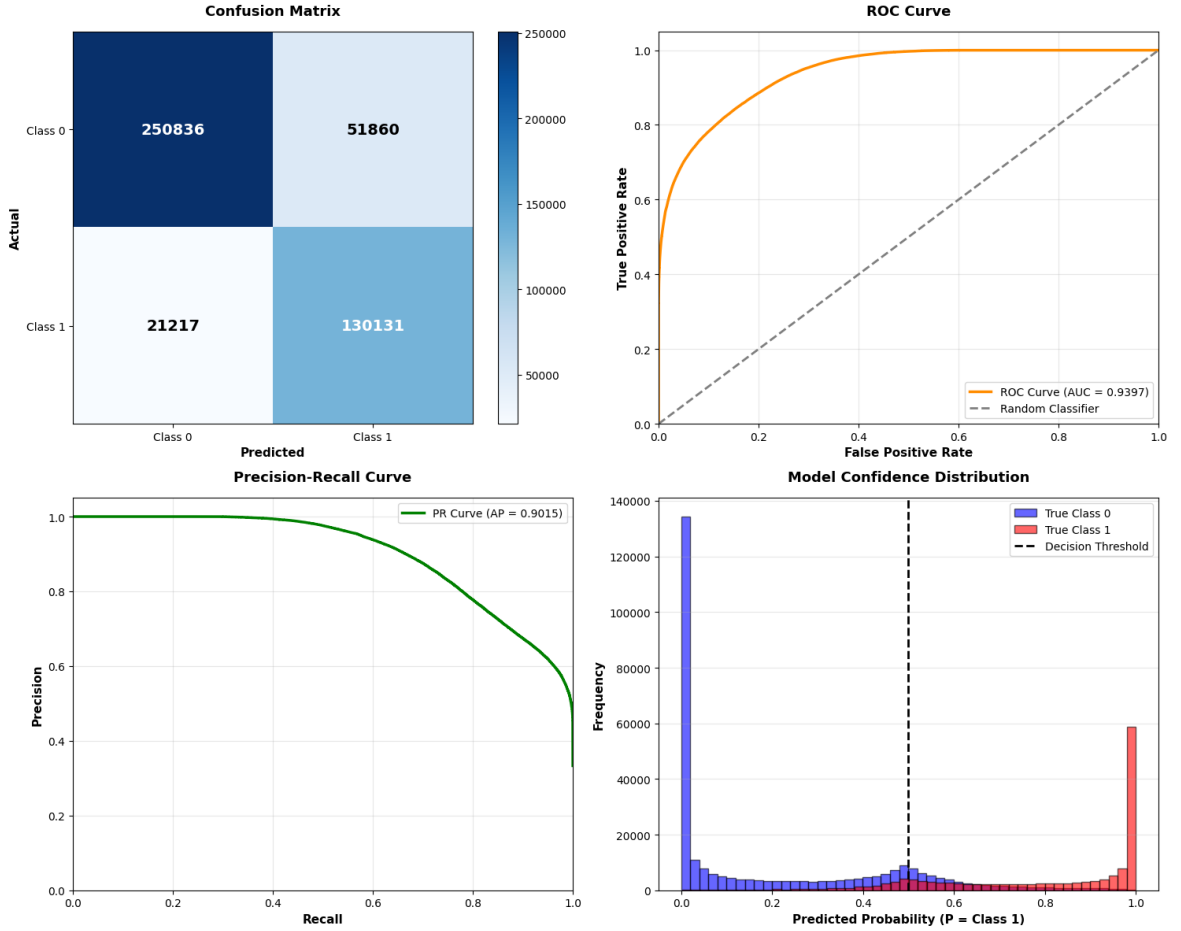


Figure 6: Comprehensive evaluation plots: (top-left) confusion matrix heatmap showing prediction patterns, (top-right) ROC curve demonstrating classification performance across thresholds, (bottom-left) precision-recall curve showing the trade-off between precision and recall, (bottom-right) distribution of predicted probabilities for each true class showing model confidence.

The ROC curve in the top-right panel demonstrated strong classification performance across all operating points, with the curve closely approaching the top-left corner (representing perfect classification). The area under this curve (AUC = 0.9397) quantified the overall discriminative ability of the model.

The precision-recall curve in the bottom-left panel showed that the model maintained reasonable precision even at high recall levels, though precision decreased somewhat as recall approached 100 percent. This trade-off is typical in classification tasks and reflects the inherent difficulty of maintaining both high precision and high recall simultaneously.

The probability distribution plot in the bottom-right panel revealed that the model produced well-separated probability distributions for the two classes, with true Class 0 samples generally receiving low predicted probabilities and true Class 1 samples receiving high predicted probabilities. Some overlap between the distributions explained the classification errors observed in the confusion matrix, but the substantial separation

indicated effective learning of discriminative features.

4.5 Class Distribution Analysis

The distribution of true labels and model predictions revealed important characteristics of the dataset and model behavior. The test set contained 302,696 samples (66.67 percent) from Class 0 and 151,348 samples (33.33 percent) from Class 1, reflecting the class imbalance present in the original data.

The model’s predictions showed a shift toward more balanced class distribution, with 272,053 samples (59.92 percent) predicted as Class 0 and 181,991 samples (40.08 percent) predicted as Class 1. This shift indicated that the weighted loss function successfully encouraged the model to pay more attention to the minority class, resulting in more frequent predictions of medial deltoid activity compared to the naive approach of simply predicting the majority class for all samples.

5 Discussion

The results of this study demonstrated that a one-dimensional convolutional neural network can effectively classify EMG signals from different deltoid muscle regions with clinically relevant accuracy. The achieved test accuracy of 83.91 percent, combined with an ROC-AUC score exceeding 0.93, suggested that automated EMG-based muscle region identification represents a viable approach for supporting clinical applications.

Several aspects of the model performance warrant detailed discussion. First, the high recall for the medial deltoid class (85.98 percent) represented a particularly important finding from a clinical standpoint. In the context of injection guidance or training systems, high recall ensures that the system reliably detects when the target muscle (medial deltoid) is active, which is crucial for confirming correct electrode placement or validating proper technique. The somewhat lower precision (71.50 percent) indicated that the model occasionally flagged activity from adjacent deltoid regions as medial deltoid activity, which could be attributed to the overlapping activation patterns that naturally occur during shoulder movements.

The choice of window size (50 samples, corresponding to 50 milliseconds of EMG data) appeared appropriate for capturing relevant temporal patterns while maintaining computational efficiency. This window duration aligned with the typical timescales of motor unit action potentials and voluntary muscle activation modulation. Alternative window sizes could be explored in future work to optimize the trade-off between temporal resolution and classification performance.

The architecture of the 1D CNN, with its three convolutional blocks of increasing depth, successfully extracted hierarchical features from the EMG signals. The rela-

tively small kernel sizes (7, 5, and 3) in successive layers allowed the network to detect patterns at multiple temporal scales, from broader activation envelopes to finer temporal fluctuations. The inclusion of batch normalization and dropout provided effective regularization, as evidenced by the model’s strong generalization to the test set.

The weighted loss function proved effective in addressing class imbalance without requiring data resampling. The resulting model demonstrated balanced performance across both classes, avoiding the common pitfall of majority class bias that often occurs in imbalanced classification problems. The shift in predicted class distribution (from 66/33 true distribution to 60/40 predicted distribution) reflected this balanced learning.

From a signal processing perspective, the preprocessing pipeline successfully extracted the EMG envelope, which provided a more stable and informative representation of muscle activity compared to raw EMG signals. The choice of filter parameters (20-450 Hz band-pass, 6 Hz envelope) followed established practices in EMG processing and effectively isolated physiologically relevant signal components.

Several limitations of this study should be acknowledged. First, the dataset comprised recordings from a single participant, which limits the generalizability of the model to diverse populations. EMG signal characteristics can vary substantially across individuals due to differences in subcutaneous tissue thickness, muscle geometry, electrode placement precision, and individual muscle activation strategies. Future work should incorporate data from multiple participants to develop more robust models.

Second, the binary classification approach (medial deltoid versus other deltoid portions) simplified the problem by combining anterior and posterior deltoid into a single class. While this approach aligned with our clinical objective, a multi-class classification framework that distinguishes all three deltoid portions individually would provide richer information and could support more comprehensive muscle activity analysis.

Third, the windowing approach with overlapping windows created temporal dependencies between adjacent samples in the dataset. While this augmentation strategy increased the effective training data size, it also meant that nearby windows were not entirely independent, which could potentially lead to overestimation of model performance. Cross-validation approaches that respect temporal structure could provide more conservative performance estimates.

Fourth, the dataset was collected during controlled dynamic movements in a laboratory setting. Real-world clinical applications would encounter more variable conditions, including different patient populations, various injection positions, potential electrode placement variations, and interference from patient movement or anxiety. Additional studies in realistic clinical settings would be necessary to validate the practical utility of such systems.

Despite these limitations, the results provide strong evidence for the feasibility of EMG-based muscle region identification. The high ROC-AUC score indicated robust

discriminative ability that could translate to practical utility even if absolute accuracy decreases somewhat in more challenging real-world conditions.

Potential applications of this technology extend beyond injection guidance. The system could support training simulators for healthcare students, providing objective feedback on electrode placement and muscle identification skills. It could also facilitate research studies investigating muscle coordination patterns or rehabilitation protocols by enabling automated, objective assessment of specific muscle activation. In sports medicine and ergonomics, similar approaches could monitor muscle recruitment patterns during athletic movements or occupational tasks.

Future research directions include several promising avenues. Expanding the dataset to include multiple participants across diverse demographics would enhance model generalizability. Incorporating additional sensor modalities, such as accelerometry or pressure sensing, could provide complementary information to improve classification accuracy. Real-time implementation on portable hardware would enable practical clinical deployment. Investigation of attention mechanisms or other advanced deep learning architectures might further improve performance. Finally, prospective clinical studies evaluating the system in realistic injection scenarios would provide essential evidence regarding practical utility and user acceptance.

6 Conclusion

This study successfully developed and validated a deep learning-based system for classifying EMG signals from different regions of the deltoid muscle, with specific focus on identifying the medial deltoid region important for intramuscular injections. The one-dimensional convolutional neural network architecture, trained on preprocessed EMG signals using a sliding window approach, achieved test accuracy of 83.91 percent with ROC-AUC score of 0.9397.

The model demonstrated particularly strong recall (85.98 percent) for detecting medial deltoid activity, a crucial characteristic for potential clinical applications. The combination of appropriate signal preprocessing, windowing strategy, CNN architecture, and class imbalance handling yielded a system capable of reliably distinguishing target muscle activity from adjacent muscle regions.

While limitations regarding single-participant data and controlled experimental conditions necessitate further validation, the results establish proof-of-concept for EMG-based muscle region identification. This technology holds promise for supporting clinical training, injection guidance systems, and various applications in rehabilitation and sports medicine. Future work incorporating diverse populations, multi-class classification, and real-world clinical validation will be essential for translating these findings into practical clinical tools.

The success of this automated approach demonstrates the broader potential of deep learning methods for analyzing physiological signals and extracting clinically relevant information. As healthcare increasingly incorporates digital technologies, systems like the one developed in this study may play important roles in enhancing safety, improving training, and supporting evidence-based clinical decision-making.

Acknowledgments

The author gratefully acknowledges Dr. Mustafa Hikmet Bilgehan Uçar for his guidance and supervision throughout this project. Thanks are also extended to the researchers who made the EMG dataset publicly available through the Zenodo repository, enabling this research.